Data Analysis, Preprocessing & Feature Engineering

Real ML work is 80% data work: load, clean, explore, transform and engineer features.

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1. Data types and data sources [Beginner]

Explanation (English)
Structured (tables), semi-structured (JSON, XML), unstructured (text, images, audio, video). Sources: CSV, SQL, APIs, scraping, logs, Kaggle, UCI, Hugging Face datasets.

Tanglish (Tamil + English) Vilakkam
Structured, semi-structured, unstructured data. Source: CSV, SQL, API, Kaggle, UCI, HF datasets.

Example / Program
```
import pandas as pd
df=pd.read_csv('https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv');print(df.head())
```

Learn more
YouTube search: Data types and data sources tutorial

2. Loading and inspecting data [Beginner]

Explanation (English)
Use head, info, describe, shape, dtypes, value_counts, nunique, isna().sum() as a first look at any dataset.

Tanglish (Tamil + English) Vilakkam
Data load panni head, info, describe, isna().sum() paarunga.

Example / Program
```
print(df.info());print(df.describe(include='all'));print(df.isna().sum())
```

Learn more
YouTube search: Loading and inspecting data tutorial

3. Handling missing values [Beginner]

Explanation (English)
Types: MCAR, MAR, MNAR. Strategies: drop, mean/median/mode imputation, forward fill, KNN/iterative imputation, missing indicator flags. Fit imputers on train only.

Tanglish (Tamil + English) Vilakkam
Missing values: drop pannalam illa mean/median/KNN-la fill pannalam. Train data-la mattum fit pannunga.

Example / Program
```
from sklearn.impute import SimpleImputer,KNNImputer
import numpy as np
X=np.array([[1,np.nan],[3,4],[5,6]])
print(SimpleImputer(strategy='median').fit_transform(X),KNNImputer().fit_transform(X))
```

Learn more
YouTube search: Handling missing values tutorial

4. Duplicates, inconsistencies, data types and string cleaning [Beginner]

Explanation (English)
Remove duplicates, fix types, standardise categories (strip, lower), parse dates, fix units.

Tanglish (Tamil + English) Vilakkam
Duplicate remove, type fix, string clean.

Example / Program
```
df=df.drop_duplicates();df['sex']=df['sex'].str.strip().str.lower();df['d']=pd.to_datetime(df['d'],errors='coerce')
```

Learn more
YouTube search: Duplicates, inconsistencies, data types and string cleaning tutorial

5. Outlier detection and treatment [Intermediate]

Explanation (English)
Detect with IQR rule, z-score, isolation forest, boxplots. Treat by capping (winsorising), transformation or removal - only if they are errors.

Tanglish (Tamil + English) Vilakkam

Outlier-a IQR/z-score-la kandupidikkalam. Error-na mattum remove pannunga.

Example / Program

```
q1,q3=df.total_bill.quantile([.25,.75]);i=q3-q1
out=df[(df.total_bill<q1-1.5*i)|(df.total_bill>q3+1.5*i)];print(len(out))
```

Learn more

YouTube search: Outlier detection and treatment tutorial

6. Exploratory Data Analysis (EDA) [Beginner]

Explanation (English)

Univariate (hist, box), bivariate (scatter, corr), multivariate (pairplot, heatmap). Look for distributions, relationships, leakage, imbalance and data-quality issues.

Tanglish (Tamil + English) Vilakkam

EDA: histogram, scatter, correlation heatmap. Data-va purinjukkradhu mukkiyam.

Example / Program

```
import seaborn as sns
sns.heatmap(df.corr(numeric_only=True),annot=True)
sns.histplot(df.total_bill,kde=True)
```

Learn more

YouTube search: Exploratory Data Analysis (EDA) tutorial

7. Pandas operations: filtering, groupby, merge, pivot, apply [Beginner]

Explanation (English)

loc/iloc, boolean masks, groupby-agg, merge/join/concat, pivot_table, melt, apply/map, window rolling/expanding, datetime features.

Tanglish (Tamil + English) Vilakkam

loc/iloc filter, groupby, merge, pivot_table.

Example / Program

```
print(df.groupby('day').agg(avg=('tip','mean'),n=('tip','size')))
print(df.pivot_table(index='day',columns='sex',values='tip',aggfunc='mean'))
```

Learn more

YouTube search: Pandas operations: filtering, groupby, merge, pivot, apply tutorial

8. Encoding categorical variables [Beginner]

Explanation (English)

Label/ordinal encoding for ordered categories, one-hot for nominal, target/mean encoding and hashing for high cardinality (beware leakage).

Tanglish (Tamil + English) Vilakkam

Category-a number-a maatha: one-hot, ordinal, target encoding.

Example / Program

```
from sklearn.preprocessing import OneHotEncoder,OrdinalEncoder
print(OneHotEncoder(sparse_output=False).fit_transform([['a'],['b'],['a']]))
```

Learn more

YouTube search: Encoding categorical variables tutorial

9. Feature scaling: standardisation, min-max, robust, log [Beginner]

Explanation (English)

Distance/gradient-based models (KNN, SVM, NN, linear) need scaled features; trees do not. StandardScaler -> mean 0 std 1; MinMax -> [0,1]; RobustScaler for outliers.

Tanglish (Tamil + English) Vilakkam

KNN, SVM, NN-ku scaling venum, trees-ku venam.

Example / Program

```
from sklearn.preprocessing import StandardScaler,MinMaxScaler
import numpy as np
X=np.array([[1.],[2.],[10.]]);print(StandardScaler().fit_transform(X).ravel(),MinMaxScaler().fit_transform(X).ravel())
```

Learn more
YouTube search: Feature scaling: standardisation, min-max, robust, log tutorial

10. Feature transformation: power, log, binning, polynomial, interaction [Intermediate]

Explanation (English)
Log/Box-Cox/Yeo-Johnson fix skew; binning discretises; polynomial & interaction features let linear models capture non-linearity.

Tanglish (Tamil + English) Vilakkam
Skewed data-ku log/PowerTransformer. Polynomial features non-linearity kudukkum.

Example / Program
```
from sklearn.preprocessing import PolynomialFeatures,PowerTransformer
import numpy as np
print(PolynomialFeatures(2).fit_transform(np.array([[2,3]])))
```

Learn more
YouTube search: Feature transformation: power, log, binning, polynomial, interaction tutorial

11. Date/time, text and geo feature engineering [Intermediate]

Explanation (English)
Extract year, month, weekday, is_weekend, lag/rolling stats; text length, TF-IDF; geo distance (haversine), clusters.

Tanglish (Tamil + English) Vilakkam
Date-la irundhu month, weekday extract. Text-la irundhu length, TF-IDF.

Example / Program
```
df['d']=pd.to_datetime(['2024-01-05','2024-03-10'])
print(df['d'].dt.month.tolist(),df['d'].dt.dayofweek.tolist())
```

Learn more
YouTube search: Date/time, text and geo feature engineering tutorial

12. Feature selection [Intermediate]

Explanation (English)
Filter (correlation, chi-square, mutual info), wrapper (RFE), embedded (Lasso, tree importance). Removes noise and speeds training.

Tanglish (Tamil + English) Vilakkam
Useless features remove pannanum: filter, wrapper, embedded.

Example / Program
```
from sklearn.feature_selection import SelectKBest,f_classif
from sklearn.datasets import load_iris
X,y=load_iris(return_X_y=True);print(SelectKBest(f_classif,k=2).fit_transform(X,y).shape)
```

Learn more
YouTube search: Feature selection tutorial

13. Dimensionality reduction: PCA, t-SNE, UMAP [Intermediate]

Explanation (English)
PCA projects onto directions of max variance (linear); t-SNE and UMAP preserve local structure for visualisation. Scale data before PCA.

Tanglish (Tamil + English) Vilakkam
PCA dimension kuraikkum. t-SNE/UMAP visualization-ku.

Example / Program

```
from sklearn.decomposition import PCA
from sklearn.datasets import load_iris
X,_=load_iris(return_X_y=True);p=PCA(2).fit(X);print(p.explained_variance_ratio_)
```

Learn more

YouTube search: Dimensionality reduction: PCA, t-SNE, UMAP tutorial

14. Train/validation/test split, stratification, leakage [Beginner]

Explanation (English)

Split BEFORE preprocessing; stratify for imbalanced classes; time-ordered split for time series; data leakage (using future/test information) silently inflates scores.

Tanglish (Tamil + English) Vilakkam

Preprocessing-ku munnadiye split pannunga. Leakage dhaan model-oda biggest enemy.

Example / Program

```
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_iris
X,y=load_iris(return_X_y=True)
Xtr,Xte,ytr,yte=train_test_split(X,y,test_size=.2,stratify=y,random_state=42)
```

Learn more

YouTube search: Train/validation/test split, stratification, leakage tutorial

15. Handling imbalanced data [Intermediate]

Explanation (English)

Use class weights, resampling (SMOTE), threshold tuning, and metrics like PR-AUC/F1 instead of accuracy.

Tanglish (Tamil + English) Vilakkam

Imbalanced data-ku accuracy vendaam, F1/PR-AUC use pannunga. class_weight, SMOTE.

Example / Program

```
from sklearn.linear_model import LogisticRegression
clf=LogisticRegression(class_weight='balanced')
```

Learn more

YouTube search: Handling imbalanced data tutorial

16. Scikit-learn Pipelines and ColumnTransformer [Intermediate]

Explanation (English)

Chain preprocessing and model so the same steps apply to train/test and avoid leakage in cross-validation.

Tanglish (Tamil + English) Vilakkam

Pipeline: preprocessing + model oru object-a. Leakage varaadhu.

Example / Program

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler,OneHotEncoder
from sklearn.linear_model import LogisticRegression
ct=ColumnTransformer([('n',StandardScaler(),['age']),('c',OneHotEncoder(handle_unknown='ignore'),['city'])])
pipe=Pipeline([('prep',ct),('m',LogisticRegression())])
```

Learn more

YouTube search: Scikit-learn Pipelines and ColumnTransformer tutorial

17. Data visualisation best practices [Beginner]

Explanation (English)

Pick the right chart (compare=bar, trend=line, distribution=hist/box, relationship=scatter), label axes, avoid misleading scales, use colour-blind safe palettes.

Tanglish (Tamil + English) Vilakkam

Correct chart select pannunga. Axis label, colour careful.

Example / Program

```
import matplotlib.pyplot as plt
fig,ax=plt.subplots();ax.bar(['A','B'],[3,5]);ax.set_ylabel('Count');plt.show()
```

Learn more

YouTube search: Data visualisation best practices tutorial

18. Data collection, labelling and data quality [Intermediate]

Explanation (English)

Label with tools (Label Studio, CVAT), measure inter-annotator agreement, document datasets (datasheets), validate with Great Expectations / pandera.

Tanglish (Tamil + English) Vilakkam

Label Studio, CVAT-la labeling. Great Expectations data quality check.

Example / Program

```
import pandera as pa
s=pa.DataFrameSchema({'age':pa.Column(int,pa.Check.ge(0))})
s.validate(pd.DataFrame({'age':[1,2]}))
```

Learn more

YouTube search: Data collection, labelling and data quality tutorial

19. Mini project: Titanic / House Prices EDA + cleaning [Beginner]

Explanation (English)

Apply everything on a public dataset: load, clean, explore, engineer features, publish notebook on Kaggle/GitHub.

Tanglish (Tamil + English) Vilakkam

Titanic / House Prices dataset-la full EDA pannunga.

Learn more

YouTube search: titanic dataset eda kaggle tutorial
Kaggle Titanic: https://www.kaggle.com/c/titanic